Code-Layout, Readability and Reusability

Date	12 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing and indentation are maintained throughout the source code for better readability.	Yes	Uses standard Python indentation (4 spaces).
2	Proper File Structure	Files and folders are logically organized into templates, static files, datasets, models, and backend scripts.	Yes	Modular project directory with separate folders.
3	Meaningful Variable Name	Variable names clearly describe their purpose and improve code readability.	Yes	Descriptive names such as wellbeing_score, user_data, and prediction_result.
4	Function / Method Names	Functions are named according to their responsibilities and follow Python naming conventions.	Yes	Functions such as load_model(), predict_wellbeing(), and preprocess_data().
5	Code Comments	Important sections of the code contain comments explaining the logic and workflow.	Yes	Comments are included for preprocessing, prediction, and visualization modules.
6	Modular Design	The application is divided into reusable modules for preprocessing, prediction, visualization, and Flask routes.	Yes	Easy to maintain and extend.
7	No Redundant Code	Duplicate or unnecessary code has been removed to improve maintainability.	Yes	Common logic is reused through functions.
8	Error Handling	Input validation and exception handling are implemented to avoid application crashes.	Yes	Handles invalid inputs, missing files, and prediction errors.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolo	yWhere Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module	Python, Pandas	Training and Prediction	High

2	Machine Learning Model	Scikit-learn	Prediction Module	High
3	Flask Routes	Flask	User Input and Result Pages	High
4	Visualization Module	Matplotlib, Seaborn	Dashboard and Prediction Results	Medium
5	HTML Templates	HTML5, Bootstrap	Multiple Web Pages	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	The project follows a modular architecture with separate folders for templates, static resources, datasets, and machine learning components.
Readability	5	Meaningful variable names, proper indentation, and clear function names improve readability.
Reusability	5	Functions and modules are reusable for training, prediction, visualization, and future enhancements.
Documentation / Comments	4	Important functions and processing steps are documented with comments. Additional API documentation can be added in future versions.
Overall Score	4.8 / 5	The application is well-structured, maintainable, reusable, and follows good coding practices suitable for an AI/ML Flask project.